

Supplementary Material

This supplementary material, accompanying the IJCAI'25 submission “Simulating Misinformation Diffusion on Social Media through CoNVaI: A Textual- and Agent-Based Diffusion Model,” details the previous study of the available datasets for the empirical evaluation, along with the full probabilities of the internal transition mechanisms of AB-SIR and CoNVaI, and the parameterization of the experimental setup for the evaluation.

1 Datasets for Empirical Evaluation

For our empirical evaluation, we aimed to recreate the observed scenarios of a news piece’s actual diffusion. Thus, we required a dataset that contained real social media interactions and other relevant components to guarantee a realistic scenario. In these terms, we needed:

- **Content.** As we have stated, shared information is an essential component of diffusion. This includes initial posts, external website links, or reactions to those messages.
- **Temporal information.** The temporal information determines when messages are shared and the users who engage with that content at any given time. The timestamps of when those messages are sent allow us to determine how the information evolves and, within the evaluation process, will enable us to determine when users engage.
- **Network.** Network topology is an essential element in the diffusion of online content. Frequently, synthetic networks are used instead, but they might not reflect the same properties as real social media networks. This is a relevant limitation since the diffusion of information relies on those connections.
- **User.** User characteristics are relevant to modeling different profiles and roles within the diffusion. Enough information on user metrics, engagement, or any other content they engage with is essential to characterize them.
- **Stance.** The stance of the users towards the information is also relevant for the evaluation, which studies diffusion in epidemiological-based terms. As such, it is crucial to distinguish between *Infected* and *Vaccinated* individuals, which is akin to their stance supporting or opposing the information.

Table 1 includes the most relevant ones we found and their characteristics. We have excluded datasets created ad-hoc since the specific tweets are not publicly available and typically require retrieving new tweets and those centered around topics unrelated to misinformation.

Only three collections of these available resources include the users’ ego networks. Out of these datasets, SNAP only contains the topology of real users, with no information on message diffusion. Similarly, MediaEval contains no information on temporal dynamics or its diffusion.

Dataset	Content	Temporal	Network	User	Stance	Topic
FakeNewsNet [Shu et al., 2020]	✓	✓	✗	✓	✗	Politics
Palin and Obama [Qazvinian et al., 2011]	✗	✓	✗	✓	✓	Politics
ReCOVery [Zhou et al., 2020]	✓	✓	✗	✓	✗	COVID-19
CoAID [Cui and Lee, 2020]	✓	✓	✗	✓	✗	COVID-19
MediaEval [Pogorelov et al., 2022]	✓	✗	✓	✗	✓	Conspiracies
PHEME-9 [Zubiaga et al., 2016]	✓	✓	✓	✓	✓	General
SNAP [Leskovec and Krevl, 2014]	✗	✗	✓	✗	✗	-

Table 1: Available datasets for the simulation of information diffusion and their main characteristics

To the best of our knowledge, only PHEME-9 contains all the pertinent data to evaluate our model under realistic information diffusion scenarios.

2 AB-SIR Transition Probabilities

This section includes the algorithm transition probabilities for the g function in the AB-SIR model. Let u_i be an agent-SIR, and let s_i, s_j be its current state and its next state, respectively. Let $m = \langle u_k, t_k, s_k \rangle$ be the current message they are reading and $U_1 \sim \mathcal{U}(0, 1)$ represent a random value drawn from a uniform distribution. The transition probabilities when $s_k = Inf$ can be expressed as follows:

$$\begin{aligned}
P(s_j = s_k \mid s_k = Inf, s_i = Neu) &= P(U_1 \leq P_{INF}) = P_{INF} \\
P(s_j = Vac \mid s_k = Inf, s_i = Neu) &= (1 - P(U_1 \leq P_{INF})) \cdot P(U_1 \leq P_{MD}) \\
&= (1 - P_{INF}) \cdot P_{MD}
\end{aligned}$$

Based on these transitions, a *Neutral* agent might turn *Infected* or *Vaccinated* after reading a message from an *Infected* agent, with the probabilities P_{INF} or $(1 - P_{INF}) \cdot P_{MD}$, respectively. The transition probabilities for the g function when $s_k = Vac$ can also be expressed as follows:

$$\begin{aligned}
P(s_j = Vac \mid s_k = Vac, s_i = Neu) &= P(U_1 \leq P_{AD}) = P_{AD} \\
P(s_j = Cu \mid s_k = Vac, s_i = Inf) &= P(U_1 \leq P_{AD}) = P_{AD}
\end{aligned}$$

After reading a message from a *Vaccinated* agent, a *Neutral* agent might turn *Vaccinated* with a probability P_{AD} , and an *Infected* agent might turn *Cured* with the same probability.

3 CoNVaI Transition Probabilities

As with AB-SIR, we include the transition probabilities of g for CoNVaI. Let u_i be an agent-CoNVaI, and let s_i, s_j be its current state and its next state, respectively. Let $m = \langle u_k, t_k, s_k \rangle$ be

the current message they are reading and let U_2 and U_3 represent two random values drawn from uniform distributions, $U_2 \sim \mathcal{U}(0, 1 - P_{nov}(p, t))$ and $U_3 \sim \mathcal{U}(0, 1 - P_{usr}(u_j) - P_{nw}(p))$, respectively. The transition probabilities when the information is unknown for u_i can be expressed as follows:

$$\begin{aligned} P(s_j = s_k \mid s_k = Inf, s_i = Neu) &= P(U_1 \leq P_{read}(u_i, t)) \cdot P(U_2 \leq P_{rpl}(p)[t]) \cdot P(U_3 \leq P_{INF}) \\ &= P_{read}(u_i, t) \cdot \frac{P_{rpl}(p)[t]}{1 - P_{nov}(p, t)} \cdot \frac{P_{INF}}{1 - P_{usr}(u_j) - P_{nw}(p)} \end{aligned}$$

$$\begin{aligned} P(s_j = Vac \mid s_k = Inf, s_i = Neu) &= P(U_1 \leq P_{read}(u_i, t)) \cdot P(U_2 \leq P_{rpl}(p)[t]) \\ &\quad \cdot (1 - P(U_3 \leq P_{INF})) \cdot P(U_3 \leq P_{MD}) \\ &= P_{read}(u_i, t) \cdot \frac{P_{rpl}(p)[t]}{1 - P_{nov}(p, t)} \cdot \left(1 - \frac{P_{INF}}{1 - P_{usr}(u_j) - P_{nw}(p)}\right) \\ &\quad \cdot \frac{P_{MD}}{1 - P_{usr}(u_j) - P_{nw}(p)} \end{aligned}$$

After assessing whether to read a message with $P_{read}(u_i, t)$, the agent determines whether the information is relevant based on $P_{nov}(p, t)$ and $P_{rpl}(p)[t]$. Based on $P_{nw}(p)$ and $P_{usr}(u_k)$, as well as the P_{INF} and P_{MD} , they would turn *Infected* or *Vaccinated*, respectively. For the case of u_i knowing the information, let $St = \{Inf, Vac\}$, and U_4 represent a random value drawn from the uniform distribution $U_4 \sim \mathcal{U}(0, 1 - P_{nov}(p, t) - P_{usr}(u_k))$. The probabilities can be expressed as follows:

$$\begin{aligned} P(s_j = s_i \mid s_k \in St, s_i = s_k) &= P(U_1 \leq P_{read}(u_i, t)) \cdot P(U_4 \leq P_{rpl}(p)[t]) \cdot P(U_1 \leq P_{OPI}) \\ &= P_{read}(u_i, t) \cdot \frac{P_{rpl}(p)[t]}{1 - P_{nov}(p, t) - P_{usr}(u_k)} \cdot P_{OPI} \end{aligned}$$

$$\begin{aligned} P(s_j = s_k \mid s_k \in St, s_i \in St \setminus \{s_k\}) &= P(U_1 \leq P_{read}(u_i, t)) \cdot P(U_4 \leq P_{rpl}(p)[t]) \cdot P(U_1 \leq P_{AD}) \\ &= P_{read}(u_i, t) \cdot \frac{P_{rpl}(p)[t]}{1 - P_{nov}(p, t) - P_{usr}(u_k)} \cdot P_{AD} \end{aligned}$$

$$\begin{aligned} P(s_j = s_i \mid s_k \in St, s_i \in St \setminus \{s_k\}) &= P(U_1 \leq P_{read}(u_i, t)) \cdot P(U_4 \leq P_{rpl}(p)[t]) \\ &\quad \cdot (1 - P(U_1 \leq P_{AD})) \cdot P(U_1 \leq P_{OPI}) \\ &= P_{read}(u_i, t) \cdot \frac{P_{rpl}(p)[t]}{1 - P_{nov}(p, t) - P_{usr}(u_k)} \cdot (1 - P_{AD}) \cdot P_{OPI} \end{aligned}$$

As before, they assess whether to read a message with a probability $P_{read}(u_i, t)$, and determine whether p is relevant based on $P_{nov}(p, t)$, $P_{usr}(u_k)$ and $P_{rpl}(p)[t]$. In such case, s_j is determined by their s_i and s_k , as well as P_{OPI} and P_{AD} .

4 Experimental Setup

The two regression components within the model have different outputs, which prompted different algorithms to be chosen. In Tables 2 and 3, we include the algorithms and hyperparameters

Algorithm	Hyperparameters	Combinations
Random Forests	n_estimators: 100, 130, 150, 180 max_depth: 10, 15, 20 min_samples_split: 3, 4, 5	816
AdaBoost	n_estimators: 100, 130, 150, 180 learning_rate: 0.01, 0.05, 0.1, 0.3, 0.5	340
Gradient Boosting	n_estimators: 100, 130, 150, 180 learning_rate: 0.05, 0.1, 0.3 max_depth: 5 min_samples_split: 2, 3, 4, 5	612

Table 2: Algorithms and Hyperparameter combinations for the *News Influence* component

Algorithm	Hyperparameters	Combinations
Decision Tree	max_depth: 10, 15, 20 min_sample_split: 3, 4, 5	153
KNeighbors	n_neighbors: 2, 5, 10, 15, 20 leaf_size_split: 5, 10, 15, 20, 30 p: 1, 2	850

Table 3: Algorithms and Hyperparameter combinations for the *Engagement Over Time* component

explored for the *News Influence* and *Engagement Over Time* models, respectively. This exploration is done through a Grid Search to cover all these combinations. Additionally, the values for the total Combinations in the tables do not include the Cross-Validation approach; they are simply the combinations of the hyperparameters with the different input data possibilities.

Regarding the final agent-based model, for each of the 58 test instances, we simulate up to $T = 1,000$ or until the diffusion stops. The combinations for the adjustable parameters for each model are contained in Table 4 and repeated for each test instance. These were selected based on our experiments and previous literature [Gausen et al., 2021, Serrano and Iglesias, 2016]. We can observe that AB-SIR has significantly fewer combinations due to its space and time complexity. The average execution time for one run was 25 minutes, while the CoNVaI model was over two minutes. The AB-SIR model could only be run on 128 GB RAM, while the CoNVaI model could run three instances simultaneously on 16 GB of RAM. With these considerations, the total AB-SIR executions average 15 hours, while CoNVaI averages almost 10. The total run time per instance is somewhat similar (although space complexity is not), which was our aim to make them comparable.

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Parameter	AB-SIR			CoNVaI		
	Min	Max	Step	Min	Max	Step
F_{NOV}	-			0.1		
F_{INFL}	-			0.1		
P_{INF}	0.05	0.15	0.05	0.05	0.15	0.05
P_{MD}	0.05	0.2	0.05	0.05	0.1	0.05
P_{AD}	0.25	0.35	0.05	0.05	0.15	0.05
P_{OPI}	-			0.1	0.25	0.05
P_{RD}	-			0.1	0.4	0.1
Combinations	36			768		

Table 4: Search space for the parameters within the AB-SIR and CoNVaI models

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